Arithmetic and complexity

Is the matrix multiplication exponent two?

Difficulty: extreme **Importance:** broadly interesting

Status: Open

Rating rationale: Extreme because closing the exponent gap is a longstanding central barrier in algebraic algorithms; broad importance follows from matrix multiplication's role throughout NLA and computational complexity.
Topic: arithmetic complexity of dense matrix multiplication

Problem statement

Over the field $\mathbb{C}$, let $M(n)$ be the minimum number of scalar additions, subtractions, multiplications, and divisions in an arithmetic straight-line program computing every entry of AB for arbitrary $A, B \in \mathbb{C}^{n\times n}$. Programs must be defined on their intended inputs. Set $\omega = \inf\{\tau : M(n) = O(n^\tau)\}$. Is $\omega = 2$? Equivalently, for every $\varepsilon > 0$, can these products be computed in $O(n^{2+\varepsilon})$ arithmetic operations? This asks about asymptotic arithmetic cost; it does not assert an $O(n^2)$ algorithm or a floating-point stability guarantee.

Why it matters

Matrix multiplication is a basic cost driver for dense NLA.

References and status

M. Bläser, *Fast Matrix Multiplication*, Graduate Surveys 5 (2013), §§1, 5, provides the computational model. E. Dupont et al., *Improving the matrix multiplication exponent with modern optimization and AlphaEvolve* (2026), abstract and §1, reports $\omega < 2.371177$, which does not reach two. Searches for "matrix multiplication exponent 2026" and "omega equals 2 proof" located improvements, not a resolution. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked Dupont et al., August 2026, and searched for exponent-two proofs and newer matrix-multiplication bounds. Its reported bound is still strictly above two, at 2.371177. No proof of exponent two or a strict lower bound above two was located. Faster finite-size identities and improved numerical optimizations of existing bounds do not decide the asymptotic equality.